

Requirement Analysis

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

Requirement Analysis identifies the functional and non-functional requirements needed to implement a secure Access Control List (ACL) solution in ServiceNow. The project ensures that only authorized users can access Institution Details records based on branch and assigned roles.

2. Functional Requirements

- Create a test user named EEE User in ServiceNow.
- Create custom roles: bb1, bb2, bb3, and bb4.
- Assign roles to the EEE User.
- Create a custom table named Institution Details (u_institution_details).
- Add fields: Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description.
- Create READ, CREATE, WRITE, and DELETE ACLs.
- Verify access using impersonation for EEE User and Admin.

3. Non-Functional Requirements

- Security: Restrict unauthorized access to records.
- Reliability: ACLs should work consistently for all users.
- Performance: Access checks should be quick and efficient.
- Scalability: More roles and branches can be added later.
- Maintainability: ACL rules should be easy to update.

4. Software and Hardware Requirements

Software Requirements:

- ServiceNow Personal Developer Instance (Australia).
- Web Browser (Google Chrome or Microsoft Edge).
- GitHub for project submission.

Hardware Requirements:

- Laptop/Desktop with internet connection.
- Minimum 4 GB RAM.

5. User Roles

Role	Responsibility
Admin	Can view, create, edit, and delete all records.

bb1	Can read only EEE branch records.
bb2	Can create new records.
bb3	Can edit EEE records.
bb4	Can delete EEE records.

6. Expected Output

The system successfully restricts record access based on user roles and branch values.

Administrators have complete access, while users receive permissions according to the ACL configuration.